

Integrated Report for Implemented Task 3, Task 4, and Task 5 Components

March 12, 2026

1 Scope, Sources, and Bundle Structure

This report consolidates the implemented scope from Task 3, Task 4, and Task 5 into one folder under `Task5/report_bundle`. The report is based on:

- the case PDF requirements for Tasks 3, 4, and 5;
- the saved notebooks in `Task3`, `Task4`, and `Task5`;
- the saved CSV, Parquet, and PNG artifacts already on disk;
- the Task 5 maintenance notes in `CONTEXT.md`, `PLAN.md`, and `ACTIVE_FIXES.md`.

No model recomputation was performed for this report. Only saved artifacts were read, summarized, copied, and packaged.

The bundle now contains:

- this report as `task5_report.tex` and `task5_report.pdf`;
- a `data/` folder with the supporting Task 3, Task 4, and Task 5 files used here;
- a `plots/` folder with the saved Task 3 figures and the copied Task 5 figures, plus Task 4 summary plots derived from the saved parquet output.

Removed problematic files. The documented problematic Task 5.9 detail files were removed from the bundle:

- `subtask_5_9_cost_carbon_summary.csv`;
- `lane_cost_carbon_2030.csv`;
- the stale 5.9 plots derived from those files.

For Task 5.9, the authoritative saved source in this report is `data/network_kpis_by_year.csv`, because that file contains the corrected delivered-unit transport KPIs. The removed detail files were flagged in `ACTIVE_FIXES.md` as problematic.

2 Task 3: Production Planning and Feasibility Scope

2.1 PDF Alignment

The implemented Task 3 scope in this repository covers:

- **Task 3.1:** estimate 99%-robust steady daily production rates by product cluster and compute how early production must start before January 1 to maintain 12-week autonomy;
- **Task 3.2:** provide a production-feasibility simulator that validates inventory dynamics under those production rates.

The repository does *not* include completed implementations for the Task 3.3 pallet layout work or the Task 3.4 truck and tractor-trailer fleet sizing work.

2.2 Task 3.1 Method

The production notebook groups models into the two PDF-defined clusters:

- Cluster 1: Floor Care, Kitchen Help, Leisure;
- Cluster 2: Safety & Security, Wall & Window Care, Exterior Care.

The method is:

1. load all Task 2 demand simulation batches;
2. aggregate annual sales by year, simulation, and cluster;
3. compute the 99th percentile annual demand for each cluster-year;
4. divide by 354 workdays per year to get the steady 99%-robust daily production rate;
5. compute the rolling 84-day forward demand requirement over the whole year;
6. locate the critical date with maximum 84-day inventory stress;
7. compute the required January 1 inventory and convert that requirement into prior start days.

The key formulas are:

$$\text{Daily Production}_{99\%} = \frac{\text{Annual Demand P99}}{354}$$

Required Inventory on Jan 1 = Max 84-day Autonomy Requirement – Net Accumulation from Jan 1 to Critical D

2.3 Task 3.1 Results

Production requirements grow substantially with market rollout. Cluster 1 rises from 47.88 units/day in 2027 to 558.06 units/day in 2034, while Cluster 2 rises from 35.31 units/day to 408.47 units/day over the same horizon.

Year	Cluster	Daily production at 99% robustness	Buffer over mean demand %
2027	Cluster 1	47.88	3.80
2027	Cluster 2	35.31	5.65
2030	Cluster 1	360.51	3.46
2030	Cluster 2	265.52	5.90
2034	Cluster 1	558.06	4.75
2034	Cluster 2	408.47	6.00

Table 1: Saved Task 3.1 production-rate outputs from `cluster_production_rates.csv`.

The prior-start calculation shows that the system needs a long prebuild phase in the early years, then stabilizes around roughly 97–120 days before year start by the later years.

Year	Cluster	Critical date	Prior start days	Required Jan 1 inventory
2027	Cluster 1	2027-11-26	171	7,917
2027	Cluster 2	2027-11-26	157	5,354
2030	Cluster 1	2030-10-09	120	41,731
2030	Cluster 2	2030-11-22	118	30,140
2034	Cluster 1	2034-10-09	109	58,969
2034	Cluster 2	2034-10-09	114	45,150

Table 2: Saved Task 3.1 prior-start outputs from `prior_start_days_corrected.csv`.

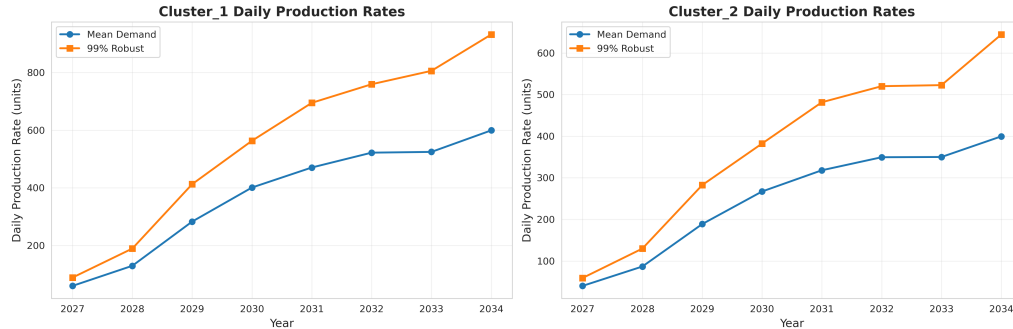


Figure 1: Saved Task 3.1 daily production rate plot by cluster and year.

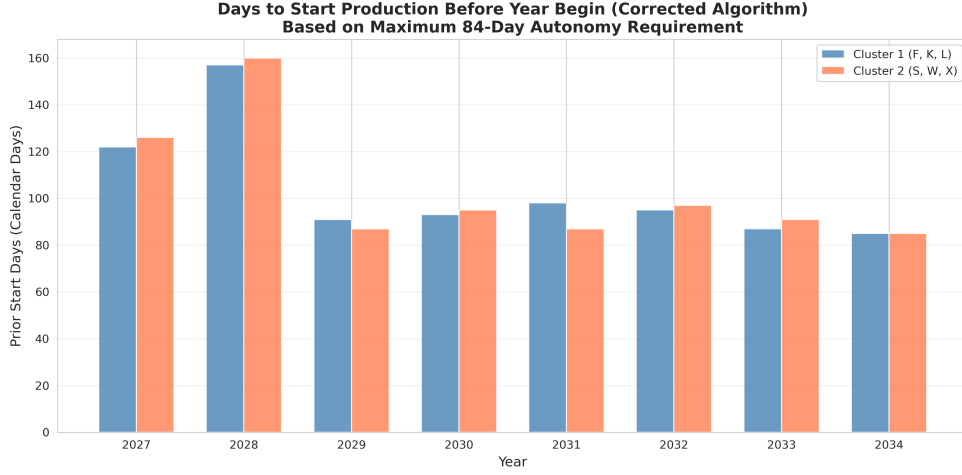


Figure 2: Saved Task 3.1 prior-start-days plot.

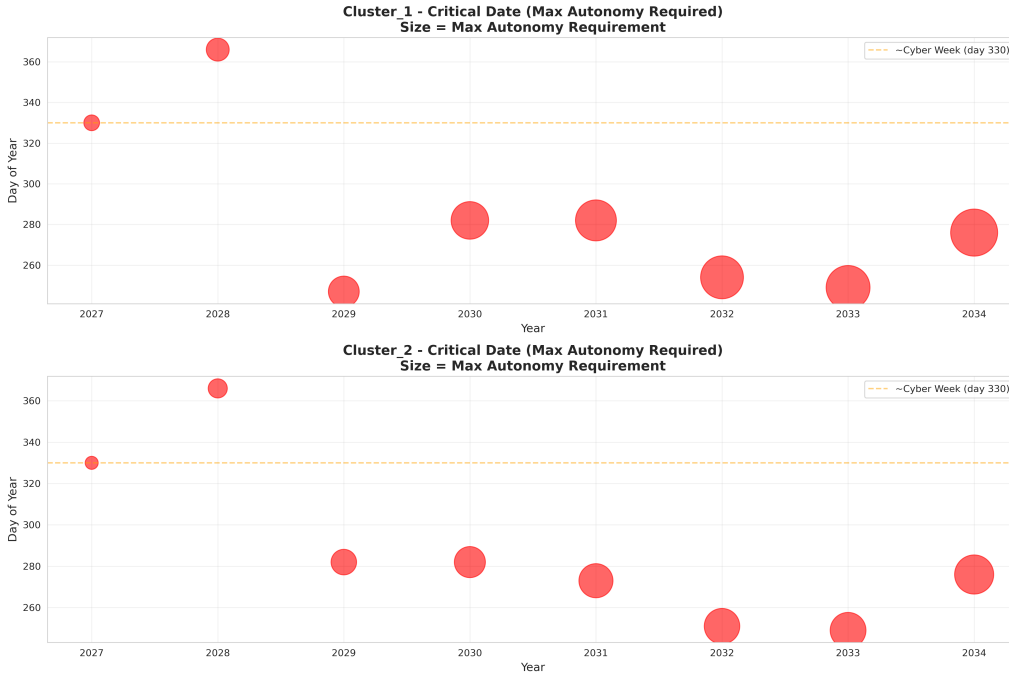


Figure 3: Saved Task 3.1 critical-date analysis. Larger markers indicate larger maximum 84-day autonomy requirements.

2.4 Task 3.2 Feasibility Simulator

The repository also includes a `Production_feasibility_simulator.ipynb` notebook for Task 3.2. Its daily logic is:

$$\text{Inventory}_{t+1} = \text{Inventory}_t + \text{Production}_t - \text{Demand}_t.$$

For each simulation and cluster, the notebook:

- starts production at the prior-start date from Task 3.1;
- produces at the steady daily 99%-robust rate on modeled workdays;
- tracks daily inventory and 84-day forward demand;
- checks whether inventory covers the next 84 days of demand.

However, the repository does not include a saved validation CSV from a completed Task 3.2 run in the packaged artifacts. The implemented simulator logic is present and reportable, but the persisted quantitative result set is limited to the Task 3.1 production and prior-start outputs.

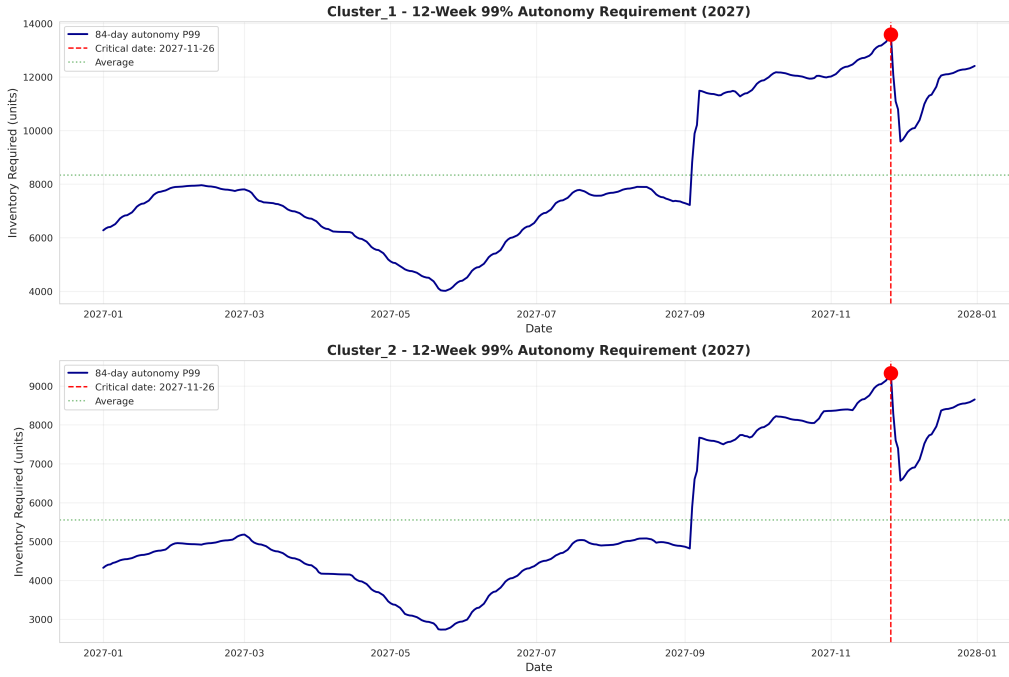


Figure 4: Saved Task 3.1 autonomy-inventory profile for 2027, used as the basis for prior-start sizing.

3 Task 4: DC-Level Demand Simulator Scope

3.1 PDF Alignment

The implemented Task 4 scope in this repository is a **DC-level daily realized demand simulator**. It supports the daily demand side of Task 4, especially the parts of the PDF that require daily sales and outbound shipment modeling from Euro DCs. It does *not* implement the full container, ship, chassis, inventory-by-site, or fleet dynamics requested across all Task 4 subtasks.

3.2 Method

The Task 4 notebook preserves the Task 2 stochastic demand logic and pushes it to DC level in four stages:

1. simulate demand at the `(country, segment)` level;
2. use population-based city proportions from Task 2 assignments;
3. collapse those city proportions into `segment_dc_weights`;
4. route each segment-day-model demand quantity directly to Euro DCs through those weights.

The final output schema is:

`(sim, date, year, euro_dc_id, model, realized_units)`.

This implementation is computationally significant because it avoids materializing the full city-level intermediate table. Instead, it uses pre-aggregated segment-to-DC weights and writes Parquet batches.

3.3 Saved Output Characteristics

The saved Task 4 run produced:

- 100 simulations over 2027–2034;
- 20 product models;
- 4 Euro DCs;
- 10 batch parquet files plus one merged `dc_daily_demand.parquet`;
- 18,992,000 final records with an 81.3% fill rate versus the dense upper bound.

Year	Average annual realized units per simulation
2027	8,574
2028	17,587
2029	46,487
2030	73,403
2031	88,753
2032	97,533
2033	105,483
2034	113,448

Table 3: Saved Task 4 average annual DC demand by year from `dc_daily_demand.parquet`.

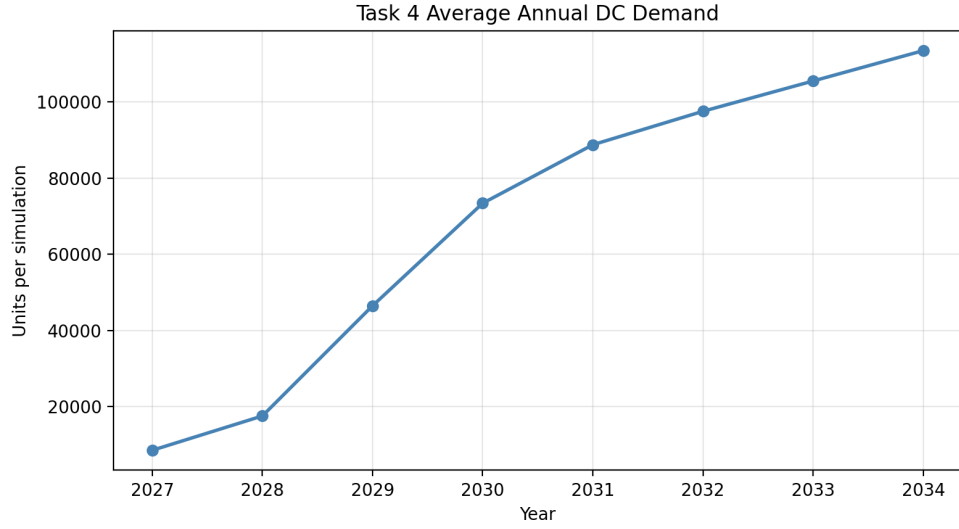


Figure 5: Task 4 annual realized demand trend derived from the saved parquet output.

By 2030 the saved DC split is already fully multi-DC, with Cologne still dominant but with material demand also routed to Madrid, Rome, and Lodz.

Year	Euro DC	Average annual units per simulation
2030	CAND_DE_koeln	27,163
2030	CAND_ES_madrid	14,781
2030	CAND_IT_rome	15,751
2030	CAND_PL_lodz	15,707
2034	CAND_DE_koeln	38,604
2034	CAND_ES_madrid	22,835
2034	CAND_IT_rome	26,926
2034	CAND_PL_lodz	25,084

Table 4: Saved Task 4 DC allocation pattern in later years.

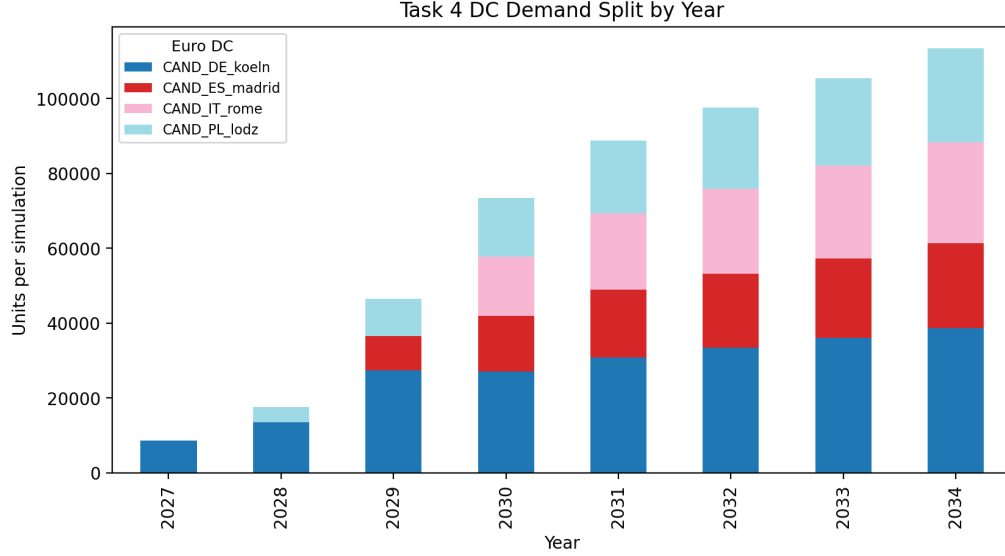


Figure 6: Task 4 DC demand mix by year derived from the saved parquet output.

Interpretation: the Task 4 artifact is best viewed as a clean daily DC-demand layer that feeds later capacity, inventory, and network work. It is not yet a full Task 4 end-to-end operational simulation of containers, tractors, chassis, ships, or site inventories.

4 Task 5: Hyperconnected PI Network Scope

4.1 Implemented Coverage

The saved Task 5 implementation covers the PI-network roadmap, PI packaging savings, route-backed OTD, route-backed realized demand, a joint-shipment illustration, relay-readiness metrics, DC sizing, transload cost, and PI profitability. The non-PI comparison requested in Task 5.14 is not part of the current scope. Task 5.13 is only partially covered through saved CO₂ outputs; there is no separate fuel or energy file in the bundle.

4.2 Algorithms and Data Flow

The Task 5 yearly pipeline runs from 2027 to 2034:

1. activate yearly open-country nodes;
2. derive relay hubs for paths exceeding the single-driver limit;
3. generate physical lane legs;
4. build shortest-elapsed-time DC-to-hub routes over those legs;
5. build a service matrix that assigns each demand node to its best PI route;
6. compute end-to-end PI OTD and purchase probability;

7. simulate daily DC demand by model;
8. size DC capacity and compute profitability.

The key Task 5 data structures are:

- **active_nodes**: the yearly node roster after opening rules and relay derivation;
- **lanes**: physical legs such as PORT_DC, DC_RELAY, and HUB_CORRIDOR;
- **routes**: reconstructed multi-leg DC-to-hub paths with elapsed time and detour;
- **service_matrix**: the node-level service view used for OTD and realized demand;
- **dc_daily_sim**: simulator-backed DC-model daily demand output;
- **kpis**: the final saved yearly KPI summary used in this report.

The Task 5 end-to-end PI OTD logic is:

$$\text{OTD} = \text{route travel elapsed} + \text{PI last mile.}$$

The saved PI last-mile assumptions follow the PDF:

- metro: 4 h for 1M+, 2 h for 250K–1M, 1 h for less than 250K;
- non-metro: 8 h within 250 km, 16 h within 500 km, 32 h beyond 500 km.

4.3 Task 5.1: Network Roadmap

The network grows from one DC in 2027 to four DCs by 2030. Coverage remains above 97% throughout the rollout, and routed detour stays tightly controlled.

Year	DCs	Peri hubs	Relay hubs	Lanes	Coverage %	Relay ready %	Mean detour
2027	1	71	0	2,622	100.00	98.09	1.001
2028	2	111	23	3,462	99.10	89.57	1.016
2029	3	232	103	6,667	97.41	74.73	1.029
2030	4	269	167	8,151	97.40	69.28	1.021
2034	4	269	167	8,151	97.40	69.28	1.021

Table 5: Saved Task 5 network KPIs from `network_kpis_by_year.csv`.

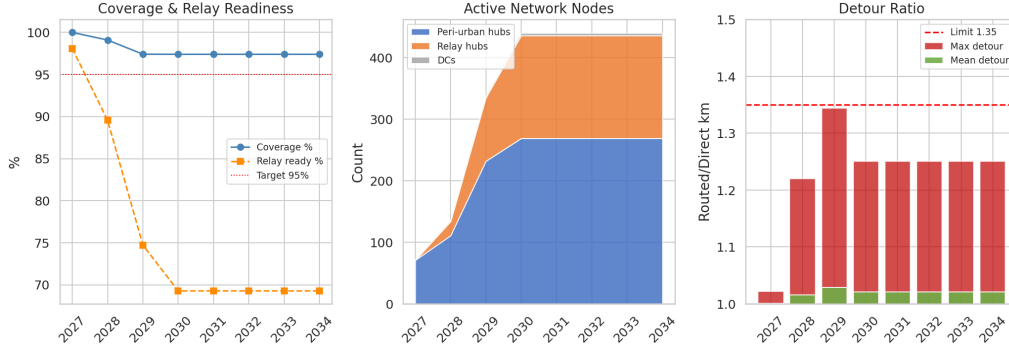


Figure 7: Saved Task 5 network KPI trends.

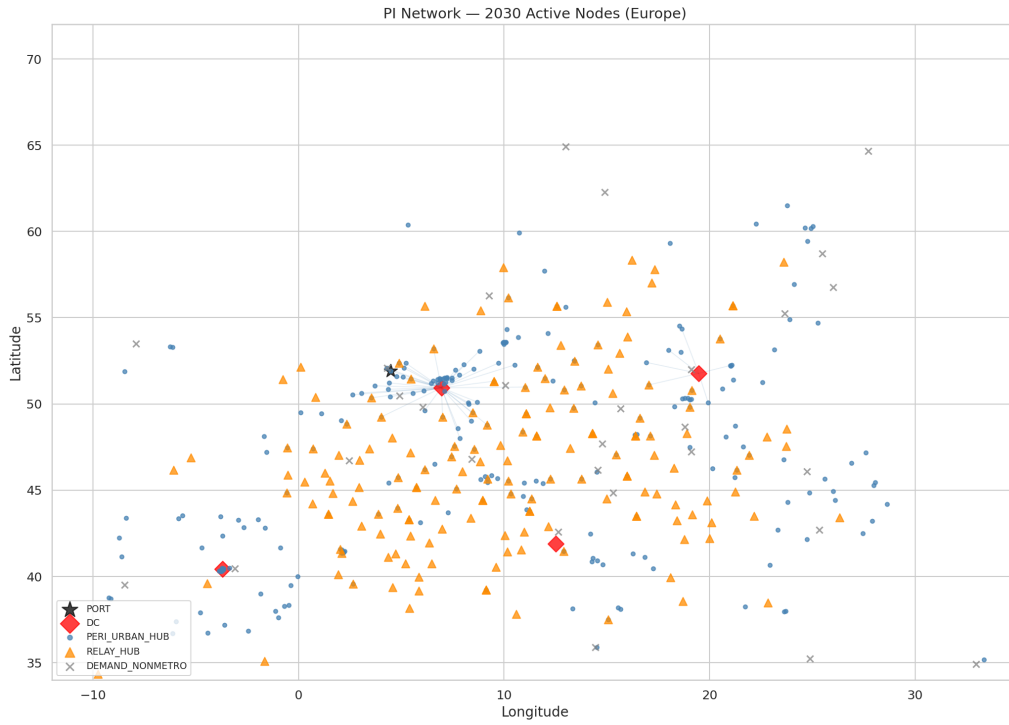


Figure 8: Saved 2030 PI network map with the port, four Euro DCs, peri-urban hubs, and relay hubs.

The saved 2030 roadmap centers the four Euro DCs on Cologne, Lodz, Madrid, and Rome. That produces a balanced west-central-east-south structure for the fully opened European market.

4.4 Task 5.2: PI Packaging and Space Savings

The saved packaging analysis shows stable structural savings:

- 33.3% fewer pallet positions;
- about 57.9% fewer TEUs;

- annual packaging plus container savings growing from about EUR 0.87M in 2027 to about EUR 10.16M in 2034.

Year	PI units	PI pallets	PI TEUs	Pallet saving %	TEU saving %	Total saving (EUR M)
2027	28,602	4,767	157	33.3	57.8	0.87
2030	218,466	36,411	1,194	33.3	57.9	6.62
2034	335,412	55,902	1,833	33.3	57.9	10.16

Table 6: Saved Task 5 PI packaging and container savings.

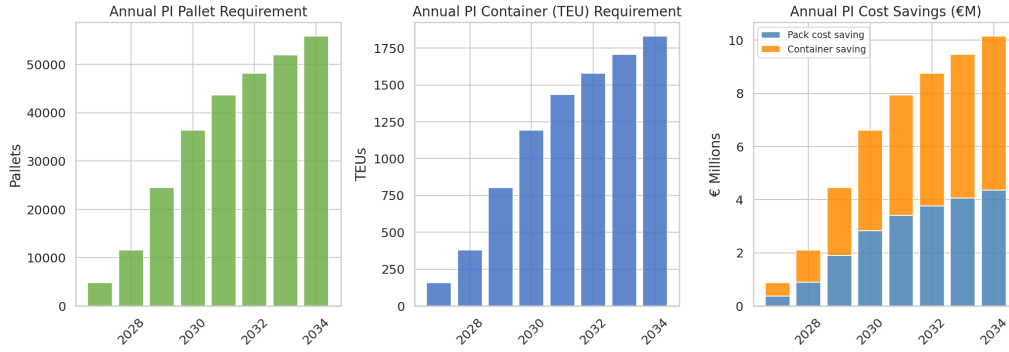


Figure 9: Saved Task 5 PI packaging and container savings plot.

4.5 Tasks 5.3 and 5.5: OTD and Realized PI Demand

The saved OTD profile shows that the largest demand volume is still in the fast service classes. The longest non-metro tail is the only segment with a substantial purchase-probability penalty.

2030 service class	Nodes	Mean OTD (h)	P95 OTD (h)	Mean purchase prob.	Annual realized units
METRO_1M+	32	14.996	30.736	0.995	31,764.6
METRO_250K-1M	127	10.029	19.797	0.999	28,386.2
METRO_<250K	108	9.612	23.889	0.996	10,117.8
METRO_UNKNOWN	6	1.000	1.000	1.000	6,233.8
NONMETRO_<=250KM	23	16.442	21.951	0.999	130,773.7
NONMETRO_<=500KM	3	31.866	34.407	0.980	7,185.1
NONMETRO_>500KM	3	62.057	78.890	0.920	4,004.7

Table 7: Saved 2030 Task 5 service-class OTD and realized demand summary.



Figure 10: Saved Task 5 OTD profile by service class.

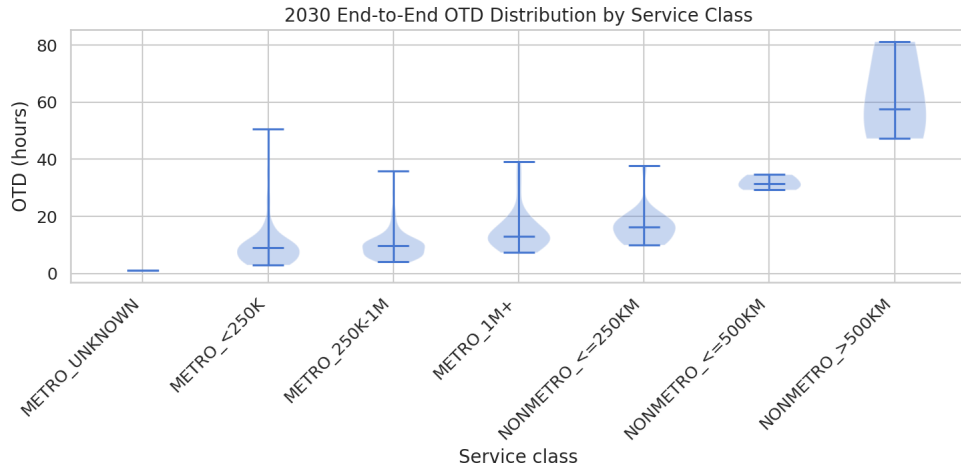


Figure 11: Saved Task 5 2030 OTD distribution by service class.

The saved PI demand summary grows from 28.6k units in 2027 to 335.4k units by 2034, with route-backed coverage staying at about 97% after the full four-DC rollout.

Year	Realized PI units	PI revenue (EUR M)	Active service nodes	Route-backed nodes	Mean purchase prob.
2027	28,602	12.97	76	76	1.0000
2030	218,466	99.10	302	293	0.9966
2034	335,411	152.14	302	293	0.9966

Table 8: Saved Task 5 PI demand summary.

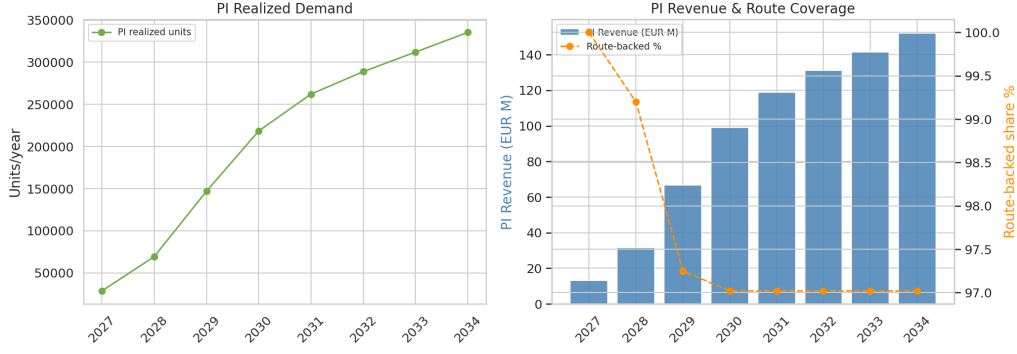


Figure 12: Saved Task 5 realized PI demand and PI revenue trend.

4.6 Task 5.4: Joint Shipment Illustration

The saved illustrative shipment path is:

PORT_NL_rotterdam → CAND_DE_koeln → METRO_BE_brussels.

The saved cumulative elapsed time is 8.635 hours:

- Rotterdam to Cologne: 203.6 km, 2.443 h drive, 2.0 h dwell;
- Cologne to Brussels: 182.7 km, 2.192 h drive, 2.0 h dwell.

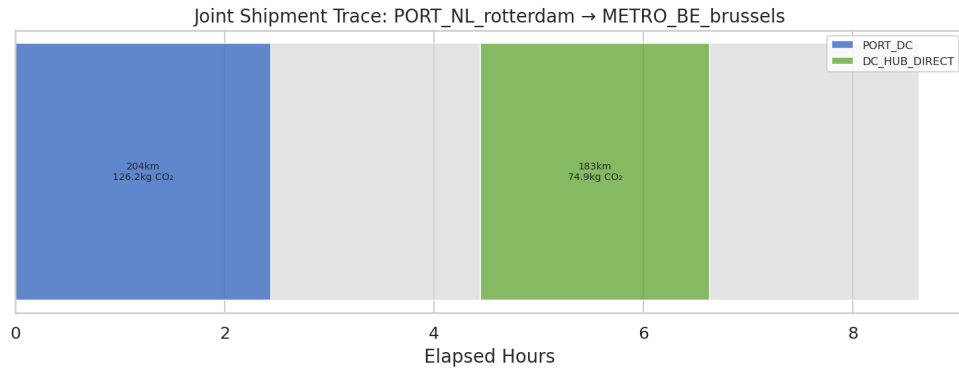


Figure 13: Saved Task 5 joint-shipment trace.

4.7 Task 5.6: Relay Readiness and Structural Autonomy

The saved Task 5.6 output operationalizes autonomy structurally through relay-readiness and dual-role-hub share rather than through explicit days-at-robustness targets.

Year	Relay readiness %	DC-DC relay required %	Dual-role hubs	Dual-role share %
2027	98.09	0.00	0	0.00
2028	89.57	0.00	56	50.45
2029	74.73	66.67	182	78.45
2030	69.28	83.33	250	92.94
2034	69.28	83.33	250	92.94

Table 9: Saved Task 5.6 relay-readiness outputs.

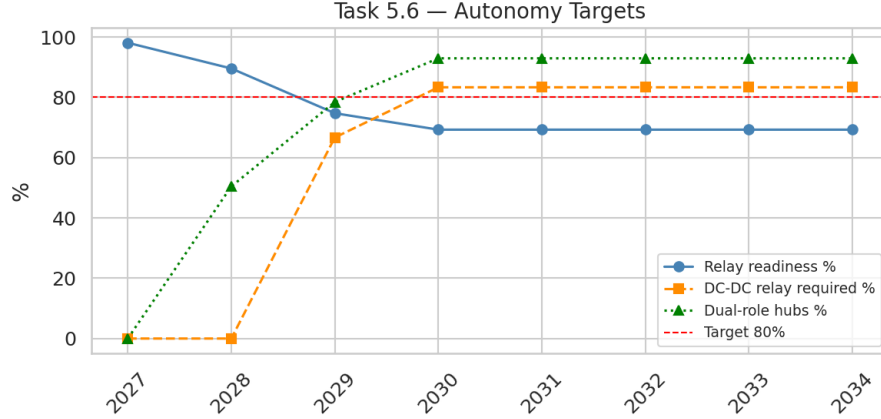


Figure 14: Saved Task 5 relay-readiness and dual-role-hub trend.

4.8 Tasks 5.9, 5.10, 5.11, and 5.12

Task 5.9. Using the corrected saved KPI file, total transport cost rises from EUR 0.76M in 2027 to EUR 24.36M in 2034. CO₂ rises from 245.6 tonnes to 11,202.8 tonnes, and the corrected delivered-unit transport cost increases from EUR 26.63/unit to EUR 72.61/unit.

Task 5.10. The saved 2030 DC sizing output is concentrated in Cologne and Lodz. Cologne carries the highest throughput requirement.

2030 DC	Annual units	P95 peak daily units	Throughput cap	Storage positions	DC cost (EUR M)
Cologne	109,711	4,747	1,584	702	1.211
Madrid	22,305	965	322	143	0.724
Rome	34,545	1,495	500	221	0.792
Lodz	51,905	2,246	750	332	0.889

Table 10: Saved Task 5 2030 DC sizing output.

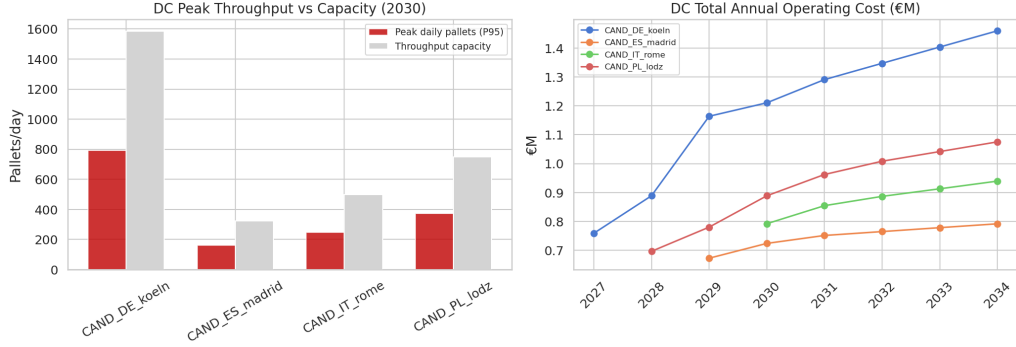


Figure 15: Saved Task 5 DC capacity and annual operating cost trend.

Task 5.11. Saved transload cost reaches EUR 655k in 2030 and EUR 1.01M in 2034.

Task 5.12. The saved PI-only profitability output remains strongly positive across all years.

Year	Transport cost (EUR M)	Total CO ₂ (t)	Cost per unit (EUR)	PI margin (EUR M)	PI margin %
2027	0.76	245.6	26.628	10.73	82.73
2030	15.13	6,901.2	69.261	75.00	75.69
2034	24.36	11,202.8	72.614	115.31	75.79

Table 11: Saved corrected Task 5 transport and profitability summary.

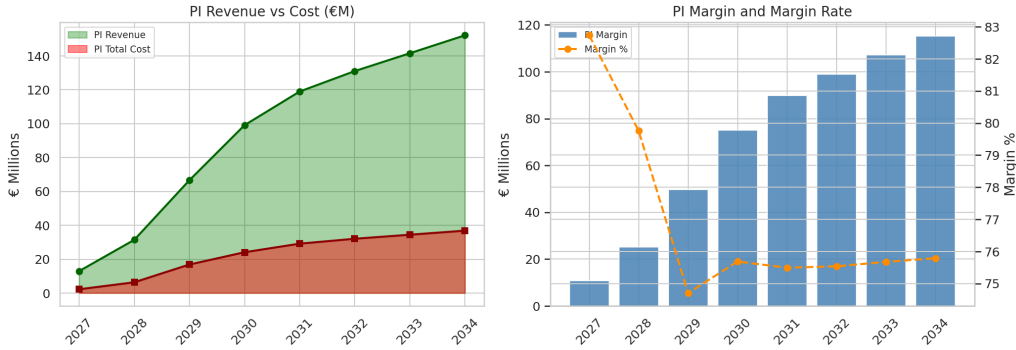


Figure 16: Saved Task 5 profitability trend.

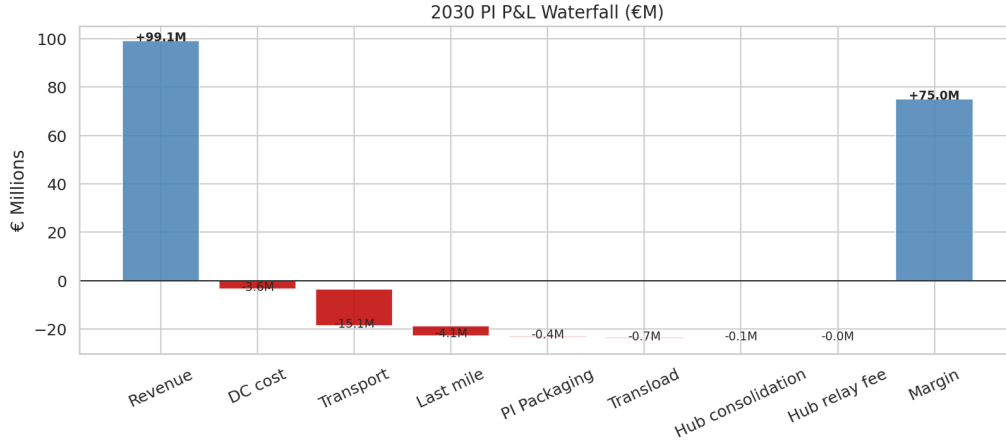


Figure 17: Saved Task 5 2030 PI profit-and-loss waterfall.

5 Discussion and Remaining Gaps

Across the implemented scope, the repository tells a coherent story:

- Task 3 provides a defensible production-rate and prior-start calculation for 12-week, 99% autonomy at the cluster level;
- Task 4 provides a scalable DC-level daily demand simulator that can feed later capacity and network work;
- Task 5 builds a route-backed PI network layer on top of that demand structure and produces credible PI-only packaging, OTD, demand, sizing, and profitability outputs.

The main limitations are:

- Task 3.2 simulator logic exists, but the bundle does not contain a saved quantitative validation result set from a completed run;
- Task 4 does not yet implement the full container, ship, chassis, inventory-by-site, and fleet dynamics requested in the PDF;
- Task 5.6 is represented by relay-readiness metrics rather than explicit autonomy targets in days and robustness percent;
- Task 5.13 has CO₂ outputs but not a separate energy estimate;
- Task 5.14 and Task 5.15 are outside the current saved scope.

6 Conclusion

The saved repository artifacts are strongest as a staged analytical chain:

- Task 3 converts uncertain demand into robust cluster-level production and launch timing;
- Task 4 converts market demand into DC-level daily realized demand;
- Task 5 converts that operational demand view into a PI-enabled European network with route-backed service, packaging savings, capacity sizing, and positive PI-only profitability.

Within that implemented scope, the results are consistent and useful. The next quality step for the project would be to close the remaining gaps around Task 3.2 persisted validation outputs, Task 4 full logistics execution, and Task 5 non-PI comparison and alternative-port analysis.